

Anna University

REGULAR EXAMINATION ? 2024

Signals and Systems ? Regular Examination, 2024

Subject Code: EC301 Semester: 3 Regulation: R23

Time: 3 Hours Max. Marks: 70

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.
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SECTION A (Unit 1: Classification of Signals)

1. Classify the given signals as energy or power signals. Calculate the energy and power of the signal. [12 Marks]
2. Describe the working principle of Classification of Signals with necessary equations. [12 Marks]
3. Explain Classification of Signals with a neat diagram/sketch and a suitable example. [12 Marks]

SECTION B (Unit 2: LTI Systems and Convolution)

4. Explain the properties of LTI systems. Determine the output of the LTI system using convolution. [12 Marks]
5. Solve a problem related to LTI Systems and Convolution. Show all the steps clearly. [12 Marks]
6. Derive the expression related to LTI Systems and Convolution and explain each term. [12 Marks]

SECTION C (Unit 3: Fourier Series and Transforms)

7. Obtain the Fourier transform of the given signal and explain the important properties of the Fourier transform. [12 Marks]
8. Describe the working principle of Fourier Series and Transforms with necessary equations. [12 Marks]
9. Discuss the engineering applications of Fourier Series and Transforms. [12 Marks]

SECTION D (Unit 4: Laplace and Z-Transforms)

10. Find the Laplace transform of the given signal with ROC. Explain the significance of the region of convergence. [12 Marks]
11. Explain the Z-transform and its relation with the Laplace transform. Solve the given difference equation using Z-transforms. [12 Marks]
12. Write short notes on Laplace and Z-Transforms. [12 Marks]

SECTION E (Unit 5: Sampling Theorem and Reconstruction)

13. State the sampling theorem. Explain aliasing and reconstruction of the sampled signal using an ideal low-pass filter. [12 Marks]

14. List the advantages and limitations of Sampling Theorem and Reconstruction. Explain

any two in detail. [12 Marks]

15. Derive the expression related to Sampling Theorem and Reconstruction and explain each term. [8 Marks]

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